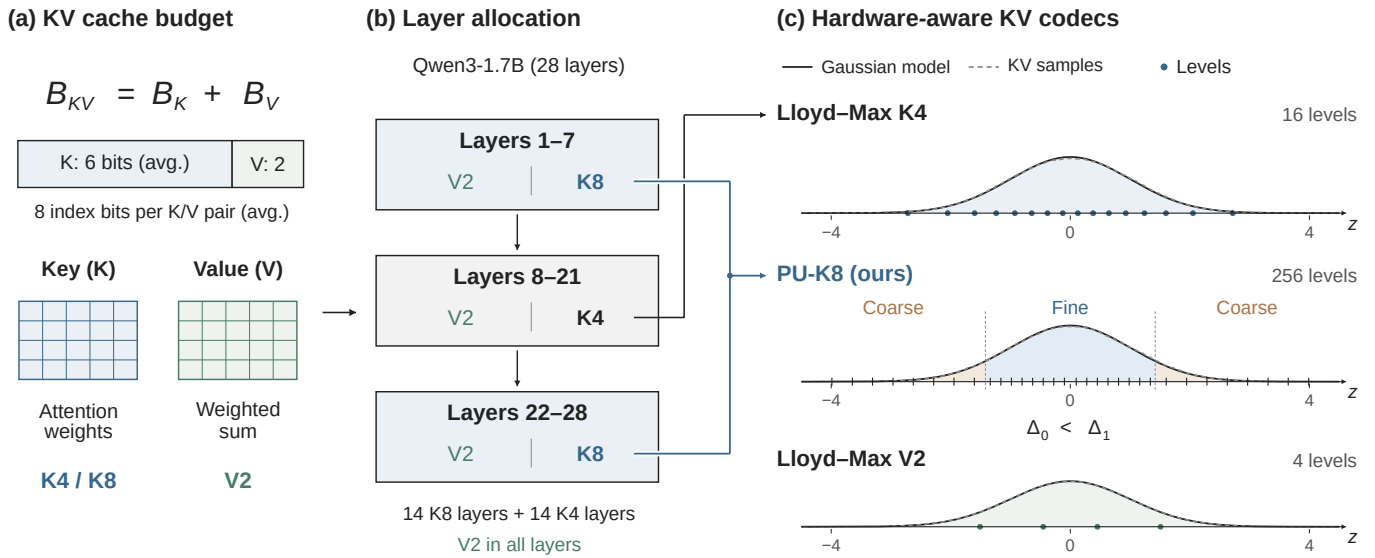
(a) KV cache budget
$B_{KV} = B_K + B_V$
K: 6 bits (avg.)
V: 2
8 index bits per K/V pair (avg.)
Key (K)
Value (V)
Attention weights
Weighted sum
K4 / K8
V2
(b) Layer allocation
Qwen3-1.7B (28 layers)
Layers 1–7
V2
K8
Layers 8–21
V2
K4
Layers 22–28
V2
K8
14 K8 layers + 14 K4 layers
V2 in all layers
(c) Hardware-aware KV codecs
Gaussian model
KV samples
Levels
Lloyd–Max K4
16 levels
−4
0
4
z
PU-K8 (ours)
256 levels
Coarse
Fine
Coarse
−4
0
4
z
$\Delta_0 < \Delta_1$
Lloyd–Max V2
4 levels
−4
0
4
z